

PRCP-1000: Portuguese Bank Marketing Campaign Analysis

1. Project Overview

This project analyzes a Portuguese bank's marketing campaign dataset to predict whether a customer will subscribe to a term deposit based on demographic, social, and economic features. The goal was to:

- Perform data analysis (EDA) to understand trends and correlations.
- Build and evaluate machine learning models for prediction.
- Provide marketing insights to improve future campaigns.

2. Data Understanding

Dataset name: bank-additional-full.csv

Records: 41,188

Features: 20 input variables + 1 target (y)

Separator: ';' (hence used `pd.read_csv('bank-additional-full.csv', sep=';')`)

Target Variable: y → indicates if a client subscribed to a term deposit (yes / no)

3. Exploratory Data Analysis (EDA)

Findings:

- Majority of customers did not subscribe to the term deposit (imbalanced dataset).
- Duration had the highest positive correlation with the target — longer calls often resulted in “yes”.
- Emp.var.rate and euribor3m showed strong relationships with campaign success.
- Job type, marital status, and education influenced customer responses.
- Used heatmap on numeric columns (age, duration, campaign) to find correlations.

4. Data Preprocessing

- Converted categorical columns using OneHotEncoder.
- Scaled numerical features using StandardScaler.
- Combined transformations using ColumnTransformer and created a Pipeline.
- Split data into train and test sets (80/20).

5. Model Building and Evaluation

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	~77%	Good	Moderate	Balanced
Random Forest	~86%	High	High	Best performer
SVM	~81%	Moderate	Moderate	Good

Best Model: svm — handled non-linear data well and gave best accuracy.

```
# SVM model i am choosing
# why : because recall : 58%
# highest recall means it finding the mpst real 'yes' customers
# stable accuracy : 81 %

# Why not Random Forest : accuracy: 86%
# high accuracy but based on 'No' class it misses true 'Yes' customers
```

6. Insights and Suggestions

- Focus on customers with higher education and stable jobs (management, technician).
- Longer and more personalized calls increase success.
- Campaigns during March–May performed better.
- Avoid excessive previous contacts — reduces conversion chance.

7. Challenges Faced

Challenge	Explanation	Solution
CSV format issue	File used ';' separator	Used sep=';' in read_csv
Heatmap error	Works only on numeric columns	Used df.select_dtypes()
Imbalanced target variable	Majority 'no' class bias	Used balanced sampling & F1-score
Categorical encoding	Many categorical columns	Used OneHotEncoder(handle_unknown='ignore')

8. Learning Outcomes

- Understood how to handle real-world banking data.
- Learned EDA, feature encoding, scaling, and correlation analysis.
- Built preprocessing and model pipelines.
- Learned model comparison and evaluation.
- Gained insights for improving marketing strategies.

9. Custom Functions And User Defined Libraries

During the project, I created several **user-defined functions** and a **custom Python library** named as {cwmeda} to automate repetitive EDA and preprocessing tasks. These functions helped improve workflow efficiency and made the analysis process more modular and reusable.

Examples of user-defined functions and modules created:

- A flexible plotting function `plot_columns()` to visualize multiple columns in grid layout with optional target variable.
- Automated EDA utilities to handle missing values, detect categorical/numeric columns, and plot correlations.
- A custom Python package (wheel file) that can be installed for automated EDA and preprocessing pipelines.
- Functions to clean data, encode categorical variables, and analyze model performance.

Purpose:

To simplify the analysis process, ensure code reusability, and make the overall pipeline more consistent and professional.

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10. Conclusion

The project successfully predicted customers likely to subscribe to term deposits. Random Forest emerged as the best-performing model. The analysis provides useful insights to improve future marketing strategies.